

YOLO Detection ONNX RUNTIME Model Inference Report

Backend Engine: ONNX Runtime

Phase 1: Configuration & System Metadata

Model Path	/workspace/Detection_Models/yolo26n.onnx
Input Dimensions	640x640
Source Images Directory	/workspace/Images
Processed Images Output	/workspace/ONNX_Output
OS	Linux 6.17.0-40-generic
Python Version	3.12.3
ONNX Runtime Version	1.27.0
Inference Device	NVIDIA RTX A4000
CUDA Version	13.0
Model Architecture	YOLO (Detect)
Total Parameters	2,408,981
File Size	10.01 MB
Number of Classes	80
Precision Mode	FP32

Model Trained Class Names:

person, bicycle, car, motorcycle, airplane, bus, train, truck, boat, traffic light, fire hydrant, stop sign, parking meter, bench, bird, cat, dog, horse, sheep, cow, elephant, bear, zebra, giraffe, backpack, umbrella, handbag, tie, suitcase, frisbee, skis, snowboard, sports ball, kite, baseball bat, baseball glove, skateboard, surfboard, tennis racket, bottle, wine glass, cup, fork, knife, spoon, bowl, banana, apple, sandwich, orange, broccoli, carrot, hot dog, pizza, donut, cake, chair, couch, potted plant, bed, dining table, toilet, tv, laptop, mouse, remote, keyboard, cell phone, microwave, oven, toaster, sink, refrigerator, book, clock, vase, scissors, teddy bear, hair drier, toothbrush

Phase 2: Inference Performance Metrics

Total Images Processed	5
Total Pipeline Time	2265.54 ms
Preprocess Latency	22.39 ms per image
Inference Latency	323.29 ms per image
Postprocess Latency	107.43 ms per image
Throughput	2.21 Images/sec
Peak Memory Usage	174.06 MB

Phase 3: Detailed Predictions Summary

Image File	Detected Objects (Confidence)	Bounding Boxes (xyxy)
Cat_November_2010-1a.jpg.webp	cat (93.5%)	[208, 142, 892, 1219]
images.jpeg	dog (94.3%)	[86, 148, 422, 514]
images1.jpeg	zebra (92.7%)	[30, 173, 587, 416]
istockphoto-1189925691-612x612.jpg	bus (90.9%)	[69, 81, 583, 350]
tigermain.jpg	zebra (95.0%)	[668, 421, 1874, 1091]